

Question Bank

Part A –

1. Explain the architecture of Django framework with suitable diagram.
 2. What is the role of `models.py` in Django? Explain with example.
 3. Differentiate between GET and POST methods in Django forms.
 4. Explain the purpose of `views.py` in Django applications.
 5. What is URL routing in Django? Explain the use of `urls.py`.
 6. Describe the steps to create a Django project and application.
 7. Explain Django Templates and their advantages.
 8. What are static files in Django? How are they managed?
 9. Explain the use of Django Admin Panel.
 10. What is migration in Django? Explain `makemigrations` and `migrate`.
 11. Explain CRUD operations in Django with suitable examples.
 12. What is ORM in Django? Explain its advantages.
 13. Explain QuerySets in Django.
 14. Differentiate between `filter()`, `get()`, and `exclude()` methods.
 15. Explain Django Form handling process.
 16. What is CSRF protection in Django? Why is it important?
 17. Explain the use of `render()` and `HttpResponse()` functions.
 18. What are Django Templates Tags and Filters? Explain with examples.
 19. Explain the use of `ImageField` in Django models.
 20. What is the purpose of `MEDIA_ROOT` and `MEDIA_URL`?
 21. Explain deployment of Django applications on Render/Heroku.
 22. What is WhiteNoise in Django deployment?
 23. Explain the concept of Aggregation in Django ORM.
 24. Explain the purpose of `aggregate()` function with example.
 25. Explain the purpose of `annotate()` function with example.
-

Part B –

1. Explain Django MTV architecture in detail with neat diagram.
2. Discuss Django Models and Database Migrations with suitable examples.
3. Explain Django Templates in detail and describe template inheritance.
4. Explain form creation and form validation in Django with example.
5. Describe the complete process of creating CRUD application in Django.
6. Explain Django ORM and QuerySets with suitable examples.
7. Discuss static files and media files management in Django.
8. Explain Django Admin customization with practical examples.
9. Explain URL mapping and dynamic URL handling in Django.
10. Explain the deployment process of a Django application on cloud platforms.

11. Discuss Django security features such as CSRF, XSS, and SQL Injection protection.
12. Explain Django authentication and authorization system.
13. Describe file upload handling in Django with example.
14. Explain Generic Views in Django and their advantages.
15. Explain the role of middleware in Django framework.
16. Discuss Django template tags and filters with examples.
17. Explain Aggregation and Annotation in Django with suitable examples.
18. Write notes on `aggregate()` and `annotate()` methods in Django ORM with examples.
19. Explain deployment challenges in Django and methods to resolve them.
20. Develop and explain a mini Django project for student management system including models, views, templates, and database operations.